

Pre-submission readiness review for the EXOhSPEC RASTI manuscript

Reference accuracy audit

I treated the submitted TeX source as the authoritative latest draft for manuscript wording, while also checking the supplied PDF where structural differences mattered. The quotation locations below refer to the submitted source text, with section and approximate page locations noted in prose.

filecite turn0file3

1. Mayor2003

Bibliographic sanity. This work exists as Mayor et al., *Setting New Standards with HARPS, The Messenger* 114, 20–24 (2003), and ESO still lists it as the reference paper for HARPS. ¹

Manuscript use. The manuscript cites it in the Introduction: “Since the first detections of exoplanets by the Doppler method \citep{Mayor2003} ... HARPS and, more recently, ESPRESSO reach sub- m s^{-1} stability ...” (Introduction opening paragraph, approx. p. 1). filecite turn0file3

What the source actually says. The HARPS paper is a 2003 commissioning/instrument paper. Its abstract says HARPS had become operational by 1 October 2003, reports commissioning-quality measurements, and describes HARPS as “currently the most precise Doppler-measurements machine in the world.” It is not a paper about the first Doppler exoplanet detections, which predate it by years. ²

Verdict. MISMATCH overall. The citation does **not** support “the first detections of exoplanets by the Doppler method.” It is only **partially relevant** to the later HARPS-stability clause.

Replacement text. *Since modern ultra-stable Doppler spectrographs such as HARPS were introduced, the field has advanced by identifying the major instrumental contributors to RV error and controlling them to stated tolerances.*

1. Pepe2021

Bibliographic sanity. This work exists as Pepe et al., *ESPRESSO at VLT. On-sky performance and first results, Astronomy & Astrophysics* 645, A96 (2021). ³

Manuscript use. It appears in the same opening Introduction sentence about HARPS/ESPRESSO reaching sub- m s^{-1} stability through vacuum enclosure, active thermal control, pressure regulation, fibre scrambling, and calibration. filecite turn0file3

What the source actually says. The abstract states that ESPRESSO was designed for ultra-high RV precision and extreme spectral fidelity; operates at $R = 140,000$ or $190,000$ on one UT and $R = 70,000$ in 4-UT mode; and demonstrates better than 25 cm s^{-1} RV precision in a single night and 50 cm s^{-1} over several months, compatible with an instrumental precision of 10 cm s^{-1} . ³

Verdict. MATCH. This is an appropriate citation for the ESPRESSO side of the stability claim.

1. Spronck2015

Bibliographic sanity. This work exists as Spronck et al., *Fiber Scrambling for High-Resolution Spectrographs. II. A Double Fiber Scrambler for Keck Observatory*, *Publications of the Astronomical Society of the Pacific* 127(956) (2015). ⁴

Manuscript use. It is used in the same opening Introduction sentence to support the phrase about HARPS/ESPRESSO combining “careful fibre scrambling.” ⁵

What the source actually says. Spronck et al. designed a double scrambler prototype for Keck/HIRES, found the spectral line-spread function 18 times more stable than slit observations and 9 times more stable than a single-fibre scrambler, and reduced RV scatter from 2.1 to 1.5 m s⁻¹. It is a good general reference for **why fibre scrambling matters**, but it is not a HARPS or ESPRESSO paper. ⁵

Verdict. PARTIAL MATCH. Relevant to the **general principle** of illumination stabilisation by fibre scrambling, but slightly misused as if it were evidence specifically about HARPS/ESPRESSO.

Replacement text. *HARPS and, more recently, ESPRESSO achieve sub-m s⁻¹ stability through aggressive environmental stabilisation and calibration, while stable illumination through fibre scrambling is a separate, well-established ingredient of precision RV design (e.g. Spronck et al. 2015).*

1. Fischer2016

Bibliographic sanity. This work exists as Fischer et al., *State of the Field: Extreme Precision Radial Velocities*, *Publications of the Astronomical Society of the Pacific* 128, 066001 (2016). ⁶

Manuscript use. It is cited first for: “The measured spectrum responds to the thermal and mechanical state of the optical train, to pressure and refractivity along the beam, and to detector behaviour, illumination stability, wavelength calibration and extraction,” and later in the Discussion in a grouped citation about what is needed before assigning an RV number. ⁷

What the source actually says. The abstract explicitly identifies improved environmental control, greater stability in the illumination of spectrometer optics, better detectors, more precise wavelength calibration, and broader bandwidth spectra as key routes to higher RV precision. ⁷

Verdict. MATCH. This is a sound broad context citation for the Introductory systems-level statement.

1. Jones2021

Bibliographic sanity. This work exists as Jones et al., *A Small Actively Controlled High-resolution Spectrograph Based on Off-the-shelf Components*, *Publications of the Astronomical Society of the Pacific* 133, 025001 (2021). ⁸

Manuscript use. The manuscript says the original EXOhSPEC paper established “a compact, off-the-shelf, fibre-fed spectrograph with $R > 70,000$ in which thorium–argon line positions were monitored in real time and a piezo-actuated element removed image drift” (Introduction, literature-context paragraph). ⁹

What the source actually says. The abstract states that the prototype is built from catalogue optical components, is fibre-fed, has resolution $> 70,000$, uses a simultaneous thorium-argon fibre for wavelength calibration, processes arc-line positions in real time, and sends corrections to piezo-electric actuators that move a mirror and remove drift. ⁸

Verdict. MATCH. This characterisation is accurate.

1. Robertson2019

Bibliographic sanity. This work exists as Robertson et al., *Ultra-Stable Environment Control for the NEID Spectrometer: Design and Performance Demonstration*, published in *Journal of Astronomical Telescopes, Instruments, and Systems* 5(1), 015003 (2019), with a matching arXiv preprint. ⁹

Manuscript use. The manuscript says the “NEID environment-control programme shows how a precision-radial-velocity instrument converts an error-budget requirement into a documented control architecture and a long-duration stability demonstration in vacuum” (Introduction, same paragraph as Jones/Blackman/Kanodia). ¹⁰

What the source actually says. The abstract says NEID’s ECS is presented in response to a $\sim 10 \text{ cm s}^{-1}$ instrumental requirement, and that full-system tests maintained vacuum below 10^{-6} Torr and RMS temperature stability better than 0.4 mK over 30 days. ¹⁰

Verdict. MATCH. This is a fair characterisation of the source.

1. Blackman2020

Bibliographic sanity. This work exists as Blackman et al., *Performance Verification of the EXtreme PREcision Spectrograph*, *The Astronomical Journal* 159, 238 (2020). ¹¹

Manuscript use. The Introduction says EXPRES “verifies performance by decomposing the instrumental error budget into environmental, mechanical, optical, calibration, fibre and detector contributions,” and the Discussion later invokes it in a grouped caution against assigning an RV precision number too early. ¹²

What the source actually says. The abstract explicitly says the paper reports a focused discussion of individual terms in the instrument error budget, including environmental, mechanical, optical, telescope/atmospheric, fibre, calibration, and detector contributions, and finds better than 10 cm s^{-1} single-measurement instrument calibration precision. ¹²

Verdict. MATCH.

1. Kanodia2023

Bibliographic sanity. This work exists as Kanodia et al., *Stable fiber-illumination for extremely precise radial velocities with NEID* (2023), available as an arXiv preprint and subsequently circulated through astronomy repositories. ¹³

Manuscript use. The manuscript says: “the NEID fibre-feed study shows that a stable environmental trace alone does not eliminate illumination-driven error” (Introduction, same instrumentation-context paragraph). ¹⁴

What the source actually says. The abstract says many parts of NEID’s optical fibre feed can lead to **uncalibratable RV errors** that cannot be corrected by a stable wavelength reference source, and that the paper presents lab and on-sky tests of illumination stability. ¹³

Verdict. PARTIAL MATCH. The source absolutely supports fibre-feed illumination as a distinct RV error term. What it does **not** literally say is the manuscript’s stronger phrasing about “a stable environmental trace alone”; that is a reasonable inference, but it is still an inference.

Replacement text. *Kanodia et al. (2023) show that illumination stability in the fibre feed is itself a distinct RV error term, with some fibre-feed errors not removable by a stable wavelength reference alone.*

1. Halverson2016

Bibliographic sanity. This work exists as Halverson et al., *A comprehensive radial velocity error budget for next generation Doppler spectrometers*, Proc. SPIE 9908, 99086P (2016). ¹⁴

Manuscript use. The manuscript says these requirements “have been consolidated into comprehensive error budgets for next-generation Doppler spectrometers” (Introduction end of paragraph), and later cites Halverson again in the Discussion’s grouped caution about RV claims. ^{filecite turn0file3}

What the source actually says. The abstract says no single source of instrumental error is expected to set the floor; instead many individual terms contribute, and the paper constructs a modular global performance budget for NEID and related instruments. ¹⁵

Verdict. MATCH.

1. Ciddor1996

Bibliographic sanity. This work exists as Ciddor, *Refractive index of air: new equations for the visible and near infrared*, *Applied Optics* 35(9), 1566–1573 (1996). ¹⁶

Manuscript use. The manuscript states: “The refractivity of air ... varies at the level of roughly 1×10^{-6} per kelvin, 3×10^{-7} per hectopascal and 1×10^{-8} per per-cent relative humidity near room conditions \cite{Ciddor1996,BonschPotulski1998}” (Introduction, physical-motivation paragraph, approx. p. 2). ^{filecite turn0file3}

What the source actually says. Ciddor provides the refractive-index equations; it does **not** tabulate those three quantities as headline coefficients. NIST’s documentation of the Ciddor and modified-Edlén calculations gives example values at 633 nm from which local room-condition sensitivities of order $-8 \times 10^{-7} \text{ K}^{-1}$, $+2.7 \times 10^{-7} \text{ hPa}^{-1}$, and $\sim 10^{-8}$ per %RH are indeed inferred. So the manuscript’s numbers are physically reasonable, but they are **derived local sensitivities**, not literal coefficients quoted by Ciddor. ¹⁷

Verdict. PARTIAL MATCH. The source supports the underlying refractivity physics, but the manuscript overstates how directly those numerical sensitivities come from the paper.

Replacement text. *Using the standard Ciddor and modified-Edlén refractivity formulae, the refractivity of air near room conditions and optical wavelengths has local sensitivities of order 10^{-6} K^{-1} , $3 \times 10^{-7} \text{ hPa}^{-1}$, and 10^{-8} per %RH, with the exact values depending on wavelength and operating point.*

1. BonschPotulski1998

Bibliographic sanity. This work exists as Bönsch & Potulski, *Measurement of the refractive index of air and comparison with modified Edlén's formulae*, *Metrologia* 35(2), 133–139 (1998), DOI 10.1088/0026-1394/35/2/8. ¹⁸

Manuscript use. Same sentence as Ciddor1996. [filecite turn0file3](#)

What the source actually says. The abstract says the authors measured dry and moist air at four visible wavelengths, adjusted the humidity dependence and the constant term of the modified Edlén dispersion formula, achieved residual reproducibility of 3×10^{-9} , and claimed a standard uncertainty of 10^{-8} for the fitted formulae at 50% RH. Again, this is support for the **formula and humidity dependence**, not for those room-condition derivative numbers as directly tabulated constants. ¹⁹

Verdict. PARTIAL MATCH. Same issue as Ciddor: correct family of sources, slightly overdirect phrasing in the manuscript.

Replacement text. Bönsch & Potulski (1998) refine the humidity dependence of the modified Edlén refractivity formula; the numerical values quoted here should therefore be described as room-condition sensitivities inferred from the standard refractivity equations, not as fixed coefficients tabulated in that paper.

1. AstromMurray2008

Bibliographic sanity. This work exists as Åström & Murray, *Feedback Systems: An Introduction for Scientists and Engineers* (Princeton University Press, 2008), with the first edition openly archived by CaltechAUTHORS. ²⁰

Manuscript use. The manuscript uses it for a generic definition of feedback/open-loop operation and for a general point that a slow nonlinear channel plus a fast discrete actuator is not well served by one continuous regulator (Introduction, control-theory paragraphs). [filecite turn0file3](#)

What the source actually says. The book is explicitly an introduction to the principles and tools for the analysis and design of feedback systems, and its front matter includes a chapter section titled “What Is Feedback?”. ²¹

Verdict. MATCH. This is a safe textbook citation for those general control concepts.

1. Ljung1999

Bibliographic sanity. This work exists as Ljung, *System Identification: Theory for the User*, 2nd ed. (Prentice Hall PTR, 1999). ²²

Manuscript use. It is cited for the claim that model-based correction “requires that the input--output relation be identified from the current record rather than transferred unchanged from an earlier state,” and again in the methods section for online re-identification. [filecite turn0file3](#)

What the source actually says. The book is a standard reference on building mathematical models of dynamic systems from observed input/output data and emphasises model validation and user choices. That comfortably supports **identification from data** and **validation in the relevant operating regime**. What it does **not** uniquely demonstrate is the manuscript's strong run-specific phrasing about the “current record” versus a prior run; that is the authors' application choice. ²³

Verdict. PARTIAL MATCH.

Replacement text. *Model-based correction here uses a system-identification step updated from measured input/output data for the operating state of the current run, following standard identification and validation practice (Ljung 1999).*

1. HoerlKennard1970

Bibliographic sanity. This work exists as Hoerl & Kennard, *Ridge Regression: Biased Estimation for Nonorthogonal Problems*, *Technometrics* 12(1), 55–67 (1970). ²⁴

Manuscript use. It is cited for using a ridge-regularised linear model and the regularised normal equations in the optical-path model. ²⁵

What the source actually says. The abstract explicitly states that the proposed estimation procedure adds small positive quantities to the diagonal of $X'X$ and yields biased estimates with smaller mean square error in nonorthogonal problems. ²⁵

Verdict. MATCH.

1. Bouchy2001

Bibliographic sanity. This work exists as Bouchy, Pepe & Queloz, *Fundamental photon noise limit to radial velocity measurements*, *Astronomy & Astrophysics* 374, 733–739 (2001). ²⁶

Manuscript use. In the Discussion it appears in a grouped citation supporting the sentence that a detector displacement is not converted into an RV uncertainty because that would require a wavelength solution, local dispersion, line-spread-function behaviour, extraction stability, calibration fidelity, and illumination analysis. ²⁷

What the source actually says. The abstract says the paper presents the methodology for computing the **fundamental photon-noise limit** to RV measurement and examines the influences of rotational broadening, spectral range, and spectral resolution. That is directly relevant to translating spectral information into an RV precision floor, but it is not a full instrument calibration-path or extraction-stability error-budget paper. ²⁷

Verdict. PARTIAL MATCH.

Replacement text. *Any conversion of detector motion into an RV uncertainty would first require a wavelength solution and local dispersion to map pixels into velocity, in addition to the broader calibration, extraction, detector, and illumination terms treated in the precision-RV instrumentation literature (e.g. Fischer et al. 2016; Blackman et al. 2020; Halverson et al. 2016).*

1. Welch1967

Bibliographic sanity. This work exists as Welch, *The Use of Fast Fourier Transform for the Estimation of Power Spectra: A Method Based on Time Averaging Over Short, Modified Periodograms*, *IEEE Transactions on Audio and Electroacoustics* 15(2), 70–73 (1967). ²⁸

Manuscript use. It is cited for the averaged-periodogram method applied to the detrended optical-path record (Methods, Safety and validity gates subsection). ²⁹

What the source actually says. Welch's abstract says the method sections the record and averages modified periodograms of the sections. ²⁹

Verdict. MATCH.

Bottom line on the references. I do **not** see hallucinated works in the bibliography list you provided. The high-priority factual problems are concentrated in **Mayor2003**, **Spronck2015**, **Ciddor1996/Bönsch & Potulski 1998**, **Kanodia2023**, **Ljung1999**, and **Bouchy2001**. The most serious of those is the plainly wrong use of **Mayor2003** for the first Doppler exoplanet detections. ³⁰

AI-generation signature and journal-readiness reassessment

The revision has helped compared with the more obviously contrast-driven style you described, but the manuscript still contains several high-salience over-smoothed markers. The risk is no longer “whole paper reads machine-written”; it is “specific sections still read heavily LLM-polished.” Those sections are the **abstract**, the **late Discussion**, the **Planned validation experiments** paragraph, and the **Conclusions**. ^{filecite turn0file3}

The most conspicuous surviving pattern is **contrastive pivoting**, especially repeated “rather than ...” framing. Representative examples include: “a sequence of methodological lessons rather than as performance benchmarks” (Introduction), “measured rather than inferred” (Introduction and Results), “rather than by design intent alone” (Discussion), “resolved-with-a-caveat rather than fully closed” (Discussion), and “rather than leaving it caveated” (Future work). The syntax is not wrong, but the recurrence is high enough to create a recognisable AI-polish cadence. Two places where this is especially noticeable are the abstract’s “the first directly measured, rather than inferred” sentence and the Discussion’s cluster of “rather than” contrasts in consecutive paragraphs. ^{filecite turn0file3}

A second surviving marker is **branded or self-conscious house language**. Phrases such as “guarded architecture,” “guarded, event-driven controller,” “self-validating philosophy,” “guarded, self-identifying, dual-bandwidth controller,” “evidence-led questions,” “load-bearing,” and “resolved-with-a-caveat” are not standard instrumentation prose; they sound like a polished explanatory voice overlaid on a technical draft. In RASTI/PASP/AJ/A&A instrumentation papers, the prose is usually plainer: one says what the controller does, what was measured, what the result was, and how the claim is bounded. The strongest surviving examples are in Discussion lines 433–435 and Conclusion line 479. ^{filecite turn0file3}

A third marker is **paragraph-end self-summary**. The clearest examples are the final sentence of the opening control-theory paragraph — “The paper addresses this through three evidence-led questions ...” — and the abstract’s final sentence about reporting the control law, environmental model, and safety logic “in full,” which tells the reader what the manuscript contains rather than adding a scientific result. That is a classic LLM-polish habit: the paragraph, after already making its point, restates its own function. ^{filecite turn0file3}

There is also still some **uniform argumentative rhythm** in the Discussion: claim, caveat, “rather than” correction, broader implication, then a final sentence that seals the interpretation. Paragraphs at lines 425–435 and 472–479 are the most patterned in that respect. Methods and results sections are much less vulnerable because the equations, thresholds, and numerical summaries pull the prose back toward domain-specific texture. ^{filecite turn0file3}

Likelihood of AI-tool flagging. My plain-language estimate is: **whole-manuscript flag risk moderate; section-level flag risk moderate-to-high in the abstract, Discussion, Future-work paragraph, and Conclusions.** A GPTZero-style detector is most likely to complain about the abstract and the late Discussion, not because the technical content is implausible, but because the local prose entropy is low and the rhetoric is unusually balanced, hedged, and self-explanatory sentence by sentence. The Methods and tabulated Results are much lower risk.

Scientific tone, claim scoping, and journal fit

On subfield norms, the manuscript is now **closer to** RASTI/PASP/AJ/A&A instrumentation style than before, but it still overshoots in several places. The strongest issue is not generic “AI-ness”; it is **register drift from measured technical prose into interpretive or promotional prose.** [filecite turn0file3](#)

The most obvious overstatement is the Figure 1 caption claim: “These observations demonstrate that environmental perturbations — particularly pressure and humidity fluctuations — are the dominant source of optical path instability in precision spectroscopic systems.” That is not justified by the captioned evidence or by the surrounding text. The paragraph immediately above says the visually distinctive feature is a **temperature** step over a 2-hour window; the broader manuscript later says the environmental model explains only about half of the optical-path variance. So this caption does more than overreach — it points in a different causal direction from the paragraph that introduces it. That should be weakened substantially or made specific to the plotted window. [filecite turn0file3](#)

The abstract slightly overstates the actuator-allocation result relative to the body’s own caveats. In the abstract, the fast stage “carries” fine-scale stabilisation “disproportionately”; in the Results and Discussion, the authors properly note that the single-actuator runs are short (2.95 h and 4.06 h) and therefore indicative rather than definitive. The body is scoped correctly; the abstract currently states the interpretation more firmly than the evidence window arguably warrants. [filecite turn0file3](#)

Two Discussion sentences still read as promotional rather than measured. The first is: “The scientific value of this work to the wider instrumentation community is that it demonstrates ...” The second is the follow-on claim that the same approach “transfers directly” to other compact or cost-constrained spectrographs “as a low-cost way” to decide whether an actuator is “earning its complexity.” That is persuasive prose, not neutral instrumentation prose. A reviewer may well agree with the intuition, but the manuscript has not demonstrated transferability across instruments; it has demonstrated one architecture on one platform. [filecite turn0file3](#)

The colloquial adjective “**load-bearing**” in “The open-loop result is the more load-bearing of the two new comparisons” is also out of register for a journal article. So are “resolved-with-a-caveat,” “genuine open-loop baseline,” and “evidence-led questions.” None is fatal alone; together they are exactly the kind of editorial polish that can make a paper feel over-composed. [filecite turn0file3](#)

One additional readiness issue is the section “**An extended architecture under validation**” followed by “**Planned validation experiments**” beginning with “*In preparation.*” This is not wrong, and RASTI can accommodate forward-looking instrumentation sections, but in a submission intended as a settled paper rather than a progress memo it weakens the centre of gravity. The present paper’s strongest contribution is the actuator decomposition plus matched open-loop comparison. The under-validation architecture material is clearly labelled, but it also lengthens the manuscript with material that is neither validated result nor fully methodological core. [filecite turn0file3](#)

Structural and cross-reference integrity

There is still a **broken figure cross-reference in the current supplied TeX source**. The Introduction says `Figure~\ref{fig:thp}` at line 71, but the figure itself is labelled `\label{fig:thp_environmental}` at line 82. If this source were compiled as supplied, that reference would not resolve. All other `\ref{}` targets in the TeX source do resolve.

filecite turn0file3

More importantly, the supplied **TeX source and supplied PDF are not the same snapshot**. The PDF I inspected has a different abstract wording and a much milder Figure 1 caption, and it shows a resolved “Figure 1” in the Introduction, whereas the TeX source still contains the label mismatch above. That means the manuscript package is not frozen to one canonical revision yet. Before submission, the authors should decide which version is authoritative and regenerate the PDF from that exact source tree.

On citations, the body uses the 16 keys you listed, and the supplied PDF does **not** show visible undefined-citation markers in the body text. However, the `.bib` file itself was not supplied here, so I could not independently audit field-level bibliography formatting or entry duplication. The key point is that the **semantic** accuracy problems are in the usage of several references, not in whether the references exist.

I did not see placeholder text of the “TK” or “XXX ref” type inside the scientific body, but “Accepted XXX. Received YYY; in original form ZZZ” is still present in the title block, which is normal for a pre-submission draft. More significant is the visible section opener “*In preparation.*” in the future-work section; that reads like an internal project-status marker rather than a polished journal section opening.

filecite turn0file3

There are also a few abstract/conclusion claims that are broader than what the body strictly establishes. The clearest is the conclusion’s statement that the actuator-decomposition method “transfers to other compact spectrographs” as a direct way of establishing actuator contribution. That may be reasonable engineering judgement, but the body presents no cross-instrument evidence. As written, it reads beyond the demonstrated scope. filecite turn0file3

Abstract deep dive

Strong instrumentation abstracts in this particular subfield follow a fairly consistent pattern. Jones et al. lead with the instrument and what was built, then give the key design features, then end on the measured performance number. Blackman et al. begin with the science/precision aim, then state exactly what testing was done, then list the measured precision and the major subsystem terms in the error budget. Robertson et al. use one sentence of science driver, then identify the subsystem being reported, then give the headline stability numbers and release/design implication. Across all three, the dominant space goes to **what was done** and **what was measured**, with numbers appearing early and carrying much of the informational load; there is very little meta-language about what the paper “gives” or how the reader should interpret the manuscript’s architecture. 31

Against that pattern, the current abstract is technically informative but structurally overburdened. It is about **266 words by count**, so slightly long for the way it is currently packed. It contains five distinct ideas: instrument/testbed context, hybrid result, single-actuator decomposition, matched open-loop result, and the image-position-versus-OPL distinction — and then adds a sixth, meta-editorial sentence about reporting the control law and safety logic “in full.” The numerical content is strong; the ordering is

the problem. Right now the abstract moves from system description to hybrid performance to actuator interpretation to open-loop comparison to OPL caveat to a final sentence about reproducibility and architecture. That is one layer too many. The closing sentence does **not** earn its place: it restates manuscript scope and uses branded wording (“guarded architecture”) rather than landing on the scientific result. [filecite0turn0file3](#)

The numbers-to-prose ratio is actually close to right for instrumentation work, but the numbers are not yet prioritised. The strongest two findings are the **matched open-loop degradation** and the **single-actuator decomposition**. Those should drive the abstract’s spine. The current version, by contrast, still gives almost equal rhetorical weight to the meta-description of the architecture and to the results themselves. That is why it feels weaker than the comparator abstracts even though it contains plenty of data.

Abstract option with actuator-decomposition emphasis

EXOhSPEC is an ambient-air, fibre-fed, high-resolution spectrograph testbed in which optical-path-length metrology, image-centroid measurements, and environmental telemetry are recorded during operation. We tested a closed-loop architecture with a slow thermo-electric correction channel and a discrete fine-optics trim stage, and separated the contribution of each actuator using matched single-actuator and open-loop comparisons on the same hardware. In a 31.44 h hybrid run, the loop remained bounded for 1604 feedback frames, with image-position scatter of 0.0635 px in X and 0.1524 px in Y , and with the radial displacement staying within 0.5 px throughout. When the fine-optics stage was disabled, the thermal channel alone degraded to 0.316 px. When the thermal channel was frozen after warm-up, the fine-optics stage alone achieved 0.138 px, matching the hybrid result with an order of magnitude fewer corrective actions. A matched open-loop interval recorded immediately after an extended hybrid run showed an eleven-fold increase in image-position scatter once both actuators were disabled. The independently monitored optical path continued to evolve during all closed-loop records, showing that bounded image position and stationary optical path are separate outcomes. These measurements identify the fast actuator as the main carrier of fine stabilisation in the present architecture while directly quantifying the benefit of the closed loop against a same-session open-loop baseline.

Abstract option with matched open-loop emphasis

We report a closed-loop image-stabilisation experiment for the ambient-air EXOhSPEC spectrograph, using synchronous image-centroid, optical-path-length, and environmental measurements to compare controlled and uncontrolled operation on the same hardware. In a 31.44 h hybrid run, the image position remained within 0.5 px of the operating reference for all 1604 feedback frames, with scatter of 0.0635 px horizontally and 0.1524 px vertically. A matched open-loop interval recorded immediately after an extended hybrid run showed an eleven-fold increase in image-position scatter when all actuation was disabled, providing a direct same-session measurement of the loop benefit. Single-actuator tests then separated the two correction channels: with the fine-optics stage disabled, the thermal channel alone degraded to 0.316 px, whereas with the thermal channel frozen after warm-up, the fine-optics stage alone delivered 0.138 px, consistent with the full hybrid result and with an order of magnitude fewer corrective actions. The independently monitored optical path continued to evolve during every closed-loop record, so the demonstrated result is bounded image-position control rather than a stationary optical path. The measurements therefore show both what the closed loop gains over open-loop operation and how that gain is distributed between the slow thermal channel and the fast fine-optics stage in the present implementation.

Abstract option with conventional context-first emphasis

High-resolution spectrographs for precise radial-velocity work are usually stabilised at the level of the full instrument, but compact ambient-air systems still need direct measurements of what closed-loop control achieves and which actuator is responsible for that gain. We address that question with EXOhSPEC, an ambient-air, fibre-fed, high-resolution spectrograph testbed that records image-centroid position, optical-path-length metrology, and environmental telemetry during operation. A closed-loop architecture combining a slow thermo-electric correction channel with a discrete fine-optics trim stage was evaluated against matched single-actuator and open-loop comparisons on the same hardware. During a 31.44 h hybrid run, the loop remained bounded for 1604 feedback frames, with scatter of 0.0635 px in X and 0.1524 px in Y , while the radial displacement stayed within 0.5 px throughout. Thermal-only operation degraded to 0.316 px, whereas fine-optics-only operation after warm-up gave 0.138 px, similar to the hybrid result and with an order of magnitude fewer corrective actions. A matched open-loop interval recorded immediately after an extended hybrid run showed an eleven-fold increase in image-position scatter when both actuators were disabled. The monitored optical path continued to evolve during all closed-loop records, demonstrating that bounded image position and stationary optical path are distinct outcomes in this system.

Final verdict

Direct verdict: not quite ready for RASTI submission yet. The paper is much closer than a first-round AI-polished draft, and the core experimental result is now much clearer, but there are still **non-negotiable factual citation fixes**, a **source/PDF snapshot inconsistency**, and a **small cluster of surviving AI-polish tonal markers** that will still attract editorial attention. My confidence in that judgement is **high**.

The highest-priority remaining fixes, ranked by impact, are these.

1. Reference-accuracy corrections that are factual and non-negotiable.

Fix the misuse of **Mayor2003** for the first Doppler exoplanet detections; narrow or recast the use of **Spronck2015** so it supports fibre scrambling in general rather than HARPS/ESPRESSO specifically; rephrase the **Ciddor1996/Bönsch & Potulski 1998** sentence so the quoted thermal/pressure/humidity sensitivities are presented as **local room-condition sensitivities inferred from the standard refractivity equations**, not as if those papers tabulate them directly; and tighten the uses of **Kanodia2023**, **Ljung1999**, and **Bouchy2001** so they support only the claims they actually make. Those are the factual issues a careful reviewer could legitimately mark in the margin. ³²

2. Residual AI-style and register problems.

The biggest remaining risk concentrations are the abstract, Discussion lines 433–435, the “In preparation” future-work paragraph, and the conclusion opening. Remove or neutralise the surviving branded phrases and editorialised pivots: “guarded,” “self-identifying,” “self-validating,” “dual-bandwidth,” “load-bearing,” “genuine open-loop baseline,” “evidence-led,” “resolved-with-a-caveat,” and repeated “rather than” turns used as rhetorical scaffolding rather than necessity. Also tone down the promotional paragraph beginning “The scientific value of this work to the wider instrumentation community is that it demonstrates ...” and the claim that the method “transfers directly” to other instruments. Those are the lines most likely to trigger “this has been over-smoothed” reactions. turn0file3

3. Abstract rewrite and manuscript freeze.

Replace the current 266-word abstract with a tighter result-led version of the kind used in Jones/Blackman/Robertson. Then freeze a **single canonical source snapshot** and regenerate the PDF from it, because the supplied TeX and supplied PDF are not identical, and the current TeX still contains the `fig:thp` / `fig:thp_environmental` mismatch. Until that is done, the manuscript is not submission-clean at the package level even apart from prose. ³¹

If those three categories are addressed — factual citations first, then residual register clean-up, then the abstract and source freeze — the paper should be in a solid position for RASTI.

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